Lecture 7 - Interpreting, Estimating, and Applying the 1st Derivative and the 2nd Derivative

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Announcements

Next week, during class time we will have our first exam! It will cover material from sections 1.1-1.7 and 3.2.3 of Active Calculus. I.e. all the lectures up to and including this Thursday's lecture.

Practice exams and the review sheet are present in the Canvas.

How to study?

- Practice, practice, and more practice! The homework problems are your friends - use them to get practice solving the relevant problems.
- Lecture Slides for review. I'll upload a compiled version of all the slides with a searchable index at the bottom so you can quickly go to the slides defining relevant terms.
- Use the practice exams as a *cumulative metric* not as an additional source of problems. Take them after you complete a round of review, actually take them in an exam-like environment (no notes, 50 minutes, uncomfy chair, etc), and use your performance on them to guide your next round of review.
- Start early! The later you start, the less time you'll have to ask me questions in office hours/via email!
- Review session - I'll hold an Exam 1 review session next Monday (from 6-8?) where you can come and ask any and all questions you have and I'll answer em!

Quiz Time!

Please get into neat rows, and spread out. Only then will I hand out the quiz. As per usual, no cheating (hopefully should go without saying).

Interpreting and Applying the First Derivative

Derivative ReCap

Last week we studied the *derivative* of a function $f(x)$:

- At a point $x = a$, we defined it as

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \tag{1}$$

- We saw that it was also equal to:
  1. The slope of the best linear approximation to $f(x)$ at $x = a$
  2. The instantaneous velocity of a particle whose position was given by $f(x)$ at $t = a$
  3. The slope of the line tangent to the graph of $y = f(x)$ at $(a, f(a))$.
- We also saw that we could construct a *derivative function*:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \tag{2}$$

  and determine some information about $f(x)$ from $f'(x)$.

Increasing-ness, Decreasing-ness, and Stationary-ness I

We can use the derivative to determine whether a function is increasing/decreasing/stationary:

Increasing Function

A function $f(x)$ is increasing around a point $x = a$ if it is increasing in *some* small neighborhood around $x = a$.

Decreasingness is similarly defined.

Increasingness via the Derivative

For a function $f(x)$ which is differentiable at $x = a$, increasingness at a is equivalent to $f'(a) > 0$. Similarly, decreasingness is equivalent to $f'(a) < 0$.

Increasingness II

Why is that true?
Idea: Because $f(x)$ is differentiable at $x = a$, it's behavior is well approximated by the best linear approximation $\mathscr{L}(x) = f(a) + f'(a)(x - a)$.
When is a linear function increasing/decreasing? When the slope (which is $f'(a)$) is positive/negative!
In terms of instantaneous velocity: positive velocity means increasing position, resp. negative velocity means decreasing position.

Graph Sketching I

We can use this idea to sketch graphs of derivative functions $f'(x)$ given graphs of $f(x)$:

And to sketch graphs of function $f(x)$ given graphs of their derivatives $f'(x)$:

Graph Sketching II

Notice how the increasingness/decreasingness of the *derivative* (e.g. the derivative of the derivative) informed us about the shape of the original function $f(x)$ - we'll see later in lecture this is due to the properties of the *second* derivative $f''(x)$.

Estimating The Derivative

Estimation I

Oftentime, in the real world, we won't have a literal function of which to take the derivative we'll instead have some (possibly large) collection of sampled datapoints.
How can we get information about the derivative of the function modeling the measured quantity from these sparse datapoints?
The derivative is a limit so:

$$\text{Estimating the derivative} \iff \text{Estimating a limit!} \tag{3}$$

Estimation II

If we're trying to estimate:

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{4}$$

We can compute the difference quotient quantities for values of $x \approx a$ or $h \approx 0$.
However, practically, there is a better choice of points about which to approximate:

Central Difference Approximation

For a collection of sampled input and output pairs (x_i, y_i), estimating the derivative of y with respect to x about $x = a$, $\frac{dy}{dx}|_{x=a}$ is best achieved by taking a pair of points (x_0, y_0) and (x_1, y_1) so that:

1. $x_0 \approx a$ and $x_1 \approx a$ (approximation requirement)
2. x_0 and x_1 are "symmetric" about $x = a$, i.e. one of them is close and slightly less than a while the other is close and slightly greater than a.

Central Difference Approximation Example

E.g.

Second and Higher Derivatives

Graph Sketching Redux

Recall that when we sketched graphs of $f(x)$ given the graph of $f'(x)$

in addition to the positivity/negativity of $f'(x)$ giving us information about the increasingness/decreasingness of $f(x)$ the increasingness/decreasingness of $f'(x)$ itself also gave us information about the curvature of $f(x)$!

The Second Derivative

The quantity that controls the increasingness/decreasingness of the *first* derivative $g(x) := f'(x)$ is *it's* derivative: $g'(x) = f''(x)$.

Second Derivative

For a function $f(x)$ that is differentiable at $x = a$ and so that $f'(x)$ is *continuous* at $x = a$ (which we say $f(x)$ is continuously differentiable at $x = a$) , the second derivative of $f(x)$ at $x = a$ denoted:

$$f''(x) \quad f^{(2)}(x) \quad \frac{d^2f}{dx^2} \quad \frac{d^2}{dx^2}f(x) \tag{5}$$

is defined to be the derivative of the function $f'(x)$ at $x = a$.
The second derivative function is defined similarly.

Quadratic Approximation

Just as the first derivative controls information about the best *linear approximation*:

$$f(x) \approx f(a) + f'(a)(x - a) \tag{6}$$

The second derivative controls information about the best *quadratic approximation*:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 \tag{7}$$

Similarly, just as the first derivative is equivalent to instantaneous *velocity* , the second derivative is equivalent to instantaneous *acceleration*!

Concavity I

$f(x) > 0 / f(x) < 0$	positivity/negativity
$f'(x) > 0 / f'(x) < 0$	increasingness/decreasingness
$f''(x) > 0 / f''(x) < 0$	?

Concavity

A function $f(x)$ is *concave up* at $x = a$ if, around $x = a$ the function is *greater* than it's best linear approximation:

$$f(x) > f(a) + f'(a)(x - a) \tag{8}$$

equivalently that it lies above it's tangent lines.
Similarly, it is *concave down* at $x = a$ if:

$$f(x) < f(a) + f'(a)(x - a) \tag{9}$$

i.e. that is lies below it's tangent lines.

Concavity II

Concavity in terms of the Second Derivative

A function $f(x)$ which is second differentiable at $x = a$ is concave up at $x = a$ if and only if $f''(a) > 0$.
Similarly it is concave down if and only if $f''(a) < 0$.

E.g.

Classifying Shapes of Functions via the First and Second Derivatives

We generically have four types of shapes a function can take given information about it's first and second derivatives:

If $f'(x) > 0$ and $f''(x) > 0$:

If $f'(x) < 0$ and $f''(x) > 0$:

If $f'(x) > 0$ and $f''(x) < 0$:

If $f'(x) < 0$ and $f''(x) < 0$:

Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!